Linux Study Notes: Package Management & Shell Scripting

1. Package Management using APT

The **APT** (Advanced Package Tool) is the primary package manager for Debian and Ubuntu-based Linux distributions. It is used to install, update, remove, and manage software packages and their dependencies.

Updating and Upgrading

Before installing new software, make sure to update your package lists to ensure you download the latest available versions:

```
sudo apt update
```

To upgrade all installed packages on your system to their newest versions:

```
sudo apt upgrade -y
```

Installing Software

To install a package, use the `install` subcommand. Below are examples of installing common utilities:

- **Install a single package:**

  ```
  sudo apt install figlet -y
  ```

- **Install multiple packages in one command:**

  ```
  sudo apt install && curl git tilix -y
  sudo apt install | curl git tilix -y
  ```

Removing Software

- **Remove a package (keeps configuration files):**

  ```
  sudo apt remove figlet
  ```

- **Purge a package (removes package and all configuration files):**

```
sudo apt purge figlet
```

- **Clean up unused dependencies:**

```
sudo apt autoremove -y
```

---

2. Creating and Running Shell Scripts

A shell script is a text file containing a sequence of commands that the shell interprets and executes.

Step 1: Create the Script File

Create a new file with a `.sh` extension using your preferred editor:

```
nano system_info.sh
```

Step 2: Add the Shebang Line and Script Commands

Every bash script must start with a **shebang** (`#!/bin/bash`) on the very first line to specify the interpreter.

```
#!/bin/bash

# Border Line
echo "————————————————————————————————————————————————————"
echo "            SYSTEM INFORMATION                      "
echo "————————————————————————————————————————————————————"

# Commands
echo -e "DATE UTC:\t$(date +'%D %T')"
echo -e "HOSTNAME:\t$(hostname)"
echo -e "UPTIME:\t\t$(uptime -p)"
echo -e "USER:\t\t$(whoami)"

# Border Line
echo "————————————————————————————————————————————————————"
```

Save the file in `nano` by pressing `Ctrl + O`, then `Enter`. Exit by pressing `Ctrl + X`.

Step 3: Make the Script Executable

New files do not have execution permissions by default. Use `chmod` to grant execute permissions:

```
chmod +x system_info.sh
```

Step 4: Run the Script

Execute the script from your current directory using `./`:

```
./system_info.sh
```

Step 5: Documenting with Screenshots

Include your verification screenshots in your Markdown report using HTML or standard markdown formatting:

```
<img src="system_info_output.png" width="700" alt="System Info Script
Execution Output">
```